

Arithmetic Sequences as Linear Functions — Finding and Fixing Off-by-One Index Errors

Almost every wrong answer in this unit comes from one habit: counting *terms* when the rule counts *steps*.

- The explicit rule is $a_n = a_1 + (n - 1)d$. The first term is already there before you move, so reaching term n takes $n - 1$ steps.
- The same sequence written as a line is $a_n = dn + (a_1 - d)$. Its constant is the value at $n = 0$, one step *before* the sequence starts — it is not the first term.
- From term p to term q inclusive there are $q - p + 1$ terms, not $q - p$. Both ends are terms.
- **The $n = 1$ test:** put $n = 1$ into any rule you write. A correct rule gives back the first term.

In each problem below: name the error in words first, then show corrected work.

1. Find and fix. An arithmetic sequence has first term $a_1 = 6$ and common difference $d = 5$. Rosa wants the 10th term and writes $a_{10} = 6 + 10(5) = 56$.

Say in one sentence why Rosa takes one step too many, then compute the correct 10th term.

Answer: _____

2. Find and fix. A sequence is described by the linear rule $a_n = 4n + 7$. Theo says “the first term is 7, because that is the constant.”

- (a) Say which term of the sequence the constant 7 really is.
(b) Compute the true first term a_1 .

Answer: _____

3. Find and fix. A list runs from the 7th term of a sequence through the 23rd term, with no terms skipped. Milo counts the terms by subtracting and reports 16 terms.

Explain what Milo's subtraction leaves out, then give the correct number of terms.

Answer: _____

4. Find and fix. An arithmetic sequence starts at 5 and increases by 6 each step. Asking which term equals 89, a student writes $5 + 6n = 89$ and reports term number 14.

(a) Write the equation the explicit rule actually gives.

(b) Solve it, and state which term of the sequence equals 89.

Answer: _____

5. Find and fix. Consider the arithmetic sequence with first term 4 and common difference 5, summed over its first 12 terms.

Priya writes $S_{12} = \frac{12}{2}(4 + [4 + 5 \cdot 12]) = 408$.

Her sum formula is right and her last term is wrong. Name the slip, then compute the correct sum.

Answer: _____

6. Find and fix. The sequence 3, 10, 17, 24, ... is to be written in the linear form $a_n = dn + b$. Dana writes $a_n = 7n + 3$ and reports $a_{20} = 143$.

(a) Test Dana's rule at $n = 1$ and say what it returns.

(b) Write the correct linear rule and compute the correct 20th term.

Answer: _____

7. Find and fix. A sequence has $a_n = 3 + 4(n - 1)$. Find the sum of the terms from a_5 through a_{20} , inclusive.

A classmate computes the two end terms correctly, then multiplies their average by $20 - 5$ and gets 735.

- (a) State how many terms the sum really contains.
- (b) Compute the correct sum.

Answer: _____

8. Explain.

- (a) Explain why the explicit rule multiplies the common difference by $n - 1$ and not by n . Refer to what the first term costs in steps.
- (b) The same sequence written as a linear function has a constant that is *not* the first term. Explain what that constant is the value of, and why a line through sequence points behaves that way.
- (c) An off-by-one index error usually produces a clean whole number, so nothing looks wrong. Describe two checks a student can run that would catch it anyway.